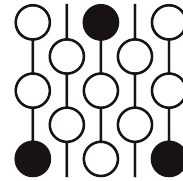


Review of Submission 7C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. This solution is entirely correct.

Novelty/exposition. All of the AI solutions for this problem laid out essentially the same proof: (1) any matching produces a graph which is a forest; (2) remove the quasi-reduced condition by noting that we can remove trivial cancelling pairs without changing $C(M)$; (3) calculate $C(M)$ for $aa^{-1}aa^{-1}aa^{-1}$; (4) show that in this case $C(M)$ is noncontractible. (This is also the approach of the human solution.)

The main differences between the AI solutions are in their level of detail in (1)–(3), and their method of showing (4). This particular solution uses a calculation of the homology group H_2 to prove (4); for what it's worth, this strikes me as the sort of bare-bones approach that a student might use in an algebraic topology problem set, while a more mature mathematician would just note that the CW complex is clearly homeomorphic to S^2 and thus not contractible. Besides this (and a minor expositional issue in the proof of Proposition 1 that I flagged there), this is quite a good proof.

2 Recommendation

Publishable with minor revisions.

3 Submission with inline comments

Abstract

We show that the complex F_w in the question is not always contractible. We give an explicit reducible quasi-reduced word w for which F_w is isomorphic to a two-dimensional cubical complex with nonzero H_2 . In fact, the relevant subcomplex is a cubical cell structure on S^2 .

3.1 Two elementary facts about matchings

Let $\mathbb{H} = \mathbb{R} \times \mathbb{R}_+$. Given a matching M , let $\Gamma(M)$ be the finite graph obtained from M by declaring the boundary points and the crossings to be vertices. Thus boundary vertices have valence 1, crossing

vertices have valence 4, and the edges of $\Gamma(M)$ are the open subarcs between consecutive vertices.

Lemma 1 (No cycles). *For every matching M satisfying the defining conditions in the question, the graph $\Gamma(M)$ is a forest.*

Proof. Suppose $\Gamma(M)$ had a cycle. Since all valence-one vertices lie on $\mathbb{R} \times \{0\}$, every cycle lies in the interior of $\mathbb{H}$. A finite plane graph with a cycle has a bounded complementary face whose boundary contains no boundary vertex on $\mathbb{R} \times \{0\}$. For the corresponding component R of $\mathbb{H} \setminus M$, some component of $M \cap \partial R$ therefore has boundary not equal to two points of $\{1, \dots, 2n\} \times \{0\}$. This contradicts the third condition in the definition of matching. $\square$

Lemma 2 (Labels around one component). *Let T be a connected component of $\Gamma(M)$. Let*

$$p_1, p_2, \dots, p_m$$

be the boundary leaves of T , listed in the cyclic order in which they occur around a small regular neighbourhood of T . Then

$$w_{p_i} = w_{p_{i+1}}^{-1}$$

for every i , with indices taken modulo m . In particular, all leaves of T are labelled by one letter ℓ and its inverse, alternating around T .

Proof. By Lemma 1, T is a tree. A sufficiently small regular neighbourhood $N(T) \subset \mathbb{H}$ is therefore a disk. Its frontier in $\mathbb{H}$ consists of arcs joining consecutive leaves p_i, p_{i+1} in the cyclic order around T . Each such frontier arc is a small push-off of a component of $M \cap \partial R$ for a complementary region R . By the defining condition on matchings, the endpoints of that component have inverse labels. Hence $w_{p_i} = w_{p_{i+1}}^{-1}$ for all i . $\square$

Corollary 1 (A uniquely occurring letter is forced). *Suppose a letter x and its inverse x^{-1} occur exactly once each in w . Then in every matching M , the two corresponding boundary points are joined by a single interval with no crossings.*

Proof. Let T be the component of $\Gamma(M)$ containing the unique x -labelled leaf. If T had more than two leaves, then that leaf would have two distinct cyclic neighbours around T , and Lemma 2 would force both of them to be labelled x^{-1} , impossible because x^{-1} occurs only once. Thus T has exactly two leaves.

Since T is a tree with m leaves and v four-valent vertices, the degree count gives

$$m + 4v = 2(m + v - 1).$$

For $m = 2$, this yields $v = 0$. Hence T is a single interval with endpoints labelled x and x^{-1} , and it has no crossings. $\square$

3.2 The counterexample

Let

$$G = \langle a, b, c \rangle$$

be the free group of rank 3, and write $\bar{a} = a^{-1}$, $\bar{b} = b^{-1}$, $\bar{c} = c^{-1}$. Consider the word

$$w = a\bar{a}b\bar{b}a\bar{a}c\bar{c}a\bar{a}.$$

This word is freely reducible to the identity. It is also quasi-reduced: its length-three subwords are

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and none has the form $\ell\ell^{-1}\ell$.

By Corollary 1, the unique occurrences of $b, \bar{b}$ are forced to form a single isolated interval, and similarly the unique occurrences of $c, \bar{c}$ are forced to form a single isolated interval. Therefore deleting those two forced intervals gives a cubical isomorphism from F_w to the matching complex for the alternating six-letter word

$$\alpha = a\bar{a}a\bar{a}a\bar{a}.$$

Conversely, any matching for α gives a matching for w by inserting small isolated intervals for $b\bar{b}$ and $c\bar{c}$ in the two gaps. This operation preserves crossings, resolutions, and attaching maps. Hence it suffices to compute the complex $K := F_\alpha$.

3.3 The complex for $a\bar{a}a\bar{a}a\bar{a}$

Label the six boundary points $1, \dots, 6$, with odd points labelled a and even points labelled $\bar{a}$. We write (ij) for an interval joining i to j , and products such as $(12)(34)(56)$ for the corresponding matching.

Because of Lemma 1, two intervals in a valid matching cross at most once. Thus an endpoint pairing determines the crossing pattern: two intervals cross exactly when their endpoints interleave. There are 15 pairings of the six endpoints. They fall into the following classes:

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The last pairing has three pairwise crossings and hence contains a cycle in the underlying graph, so it is invalid by Lemma 1. The pairings in the first three rows satisfy the side-label condition. For the two-crossing pairings, the sides of a regular neighbourhood connect the cyclically consecutive pairs

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Thus K has five vertices. Name them

$$N = (12)(34)(56), \quad S = (16)(23)(45),$$

and

$$E_1 = (12)(36)(45), \quad E_2 = (14)(23)(56), \quad E_3 = (16)(25)(34).$$

The six edges are precisely the edges joining each of N, S to each E_i . The three square two-cells are:

$$D_{12} = (13)(25)(46), \quad D_{13} = (14)(26)(35), \quad D_{23} = (15)(24)(36),$$

with boundaries

$$\partial D_{12} = NE_1SE_2N,$$

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There are no higher-dimensional cells because a valid matching on three intervals has at most two crossings by Lemma 1.

Proposition 1. *The complex K has nonzero second homology.*

Proof. Orient the edges a_i from N to E_i , and orient the edges b_i from E_i to S . Orient the square D_{ij} by the boundary path

$$N \longrightarrow E_i \longrightarrow S \longrightarrow E_j \longrightarrow N.$$

Then

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Therefore

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$$\partial z = (a_1 + b_1 - b_2 - a_2) - (a_1 + b_1 - b_3 - a_3) + (a_2 + b_2 - b_3 - a_3) = 0.$$

Since K has no 3-cells, this nonzero cellular 2-cycle represents a nonzero class in $H_2(K; \mathbb{Z})$.

Reviewer 1

Minor comment: this proof makes complete intuitive sense and would probably be unremarkable if written by a human, but I note that the proof implicitly assumes that we're using the cellular model for homology as opposed to say the singular model. However, this isn't stated anywhere except for a single passing use of the word "cellular" at the end. Given how careful the rest of the proof is, I found this omission a bit surprising.

□

Theorem 1. *The complex F_w is not contractible for*

$$w = a\bar{a}b\bar{b}a\bar{a}c\bar{c}a\bar{a}.$$

Consequently, the answer to the question is negative in general.

Proof. As shown above, $F_w \cong K$. By Proposition 1,

$$H_2(F_w; \mathbb{Z}) \cong H_2(K; \mathbb{Z}) \neq 0.$$

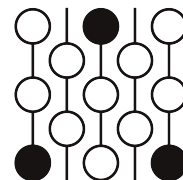
A contractible CW complex has trivial reduced homology in all positive degrees. Hence F_w is not contractible. $\square$

Review of Submission 7C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

The most difficult part of the problem is deciding whether the statement is true or false. AI correctly determined that it is false and found a valid counterexample. I have included some minor comments below, but the proof is acceptable for publication without changes.

Correctness. The proof is correct.

Novelty. The proof is not particularly novel. The counterexample given is the same as the one given in the human solution, and very similar to other AI generated solutions.

Exposition. None of the proofs included pictures, which would appear in most human solutions.

2 Recommendation

The proof is essentially flawless.

3 Submission with inline comments

Abstract

We show that the complex F_w in the question is not always contractible. We give an explicit reducible quasi-reduced word w for which F_w is isomorphic to a two-dimensional cubical complex with nonzero H_2 . In fact, the relevant subcomplex is a cubical cell structure on S^2 .

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be the boundary leaves of T , listed in the cyclic order in which they occur around a small regular neighbourhood of T . Then

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for every i , with indices taken modulo m . In particular, all leaves of T are labelled by one letter ℓ and its inverse, alternating around T .

Reviewer 2

If one is very pedantic: the tree leaves lie in T and do not "occur around a small regular neighbourhood of T ".

Proof. By Lemma 1, T is a tree. A sufficiently small regular neighbourhood $N(T) \subset \mathbb{H}$ is therefore a disk. Its frontier in $\mathbb{H}$ consists of arcs joining consecutive leaves p_i, p_{i+1} in the cyclic order around T . Each such frontier arc is a small push-off of a component of $M \cap \partial \bar{R}$ for a complementary region R . By the defining condition on matchings, the endpoints of that component have inverse labels. Hence $w_{p_i} = w_{p_{i+1}}^{-1}$ for all i . $\square$

Reviewer 2

Boundary is a more standard name for frontier. As above, if the arcs are actually part of $\partial N(T)$, then they cannot join leaves which lie in the interior of $N(T)$.

Corollary 1 (A uniquely occurring letter is forced). *Suppose a letter x and its inverse x^{-1} occur exactly once each in w . Then in every matching M , the two corresponding boundary points are joined by a single interval with no crossings.*

Proof. Let T be the component of $\Gamma(M)$ containing the unique x -labelled leaf. If T had more than two leaves, then that leaf would have two distinct cyclic neighbours around T , and Lemma 2 would force both of them to be labelled x^{-1} , impossible because x^{-1} occurs only once. Thus T has exactly two leaves.

Since T is a tree with m leaves and v four-valent vertices, the degree count gives

$$m + 4v = 2(m + v - 1).$$

For $m = 2$, this yields $v = 0$. Hence T is a single interval with endpoints labelled x and x^{-1} , and it has no crossings. $\square$

It is clear to a human that a tree with 2 leaves is an interval. The formal justification of that fact in the second paragraph of the proof is fine, but it assumes the knowledge that a tree with n vertices has $n - 1$ edges. This is indeed well-known, but slightly less obvious than the statement that is being proven. So I would remove the last paragraph entirely.

3.2 The counterexample

Let

$$G = \langle a, b, c \rangle$$

be the free group of rank 3, and write $\bar{a} = a^{-1}$, $\bar{b} = b^{-1}$, $\bar{c} = c^{-1}$. Consider the word

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Reviewer 2

It would be great if the solution included pictures.

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Since K has no 3-cells, this nonzero cellular 2-cycle represents a nonzero class in $H_2(K; \mathbb{Z})$. $\square$

A human would probably note that the resulting complex is homeomorphic to S^2 rather than compute homology.

Theorem 1. *The complex F_w is not contractible for*

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Consequently, the answer to the question is negative in general.

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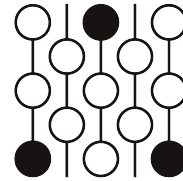
A contractible CW complex has trivial reduced homology in all positive degrees. Hence F_w is not contractible. $\square$

Review of Submission 7C

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. The solution is correct.

Novelty. The solution understands the novel definition, correctly identifies that a related graph has to be a forest, and realizes that one can insert/delete spurious additional letter-inverse letter pairs. Among the solutions that I read, this was one of the top two well-written ones—most like a mathematician.

2 Recommendation

minor revisions; it will be nice to have a bit more justification for one sentence.

3 Submission with inline comments

Abstract

We show that the complex F_w in the question is not always contractible. We give an explicit reducible quasi-reduced word w for which F_w is isomorphic to a two-dimensional cubical complex with nonzero H_2 . In fact, the relevant subcomplex is a cubical cell structure on S^2 .

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needs further justification that the endpoint pairing fixes the isotopy type of M

Reviewer 3

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